An Integer Linear Programming Model Based on Competences for Student-Industry Placement Allocation

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ABSTRACT

Allocating students to industry placements is a necessary task in educational contexts that need optimization techniques. This article proposes a novel integer linear programming model where, differently to other proposals, students' and businesses' preferences are represented by means of competences. The model has been evaluated by using commercial and non-commercial solvers, showing the need for alternative open algorithms for larger instances.

CCS CONCEPTS

• **Applied computing** → *Operations research*; **Education**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Student-industry placement allocation, competences, preferences, optimization, integer linear programming

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1 INTRODUCTION

In educational contexts, it is common for students to undertake placements in companies during their training. This type of training is usually seen as beneficial for both companies and students, as companies do not incur significant costs, and students can gain valuable professional experience for their future. Currently, educational institutions manually assign students to companies, which entails much human effort and difficulty in finding the optimal solution. In this regard, several works in the literature attempt to propose optimization models for this problem, which is known as student-industry placement allocation (SIPA) problem. For example, [1] presents an optimization algorithm that considers the distance lecturers traveled to supervise students'. As another example, [2] tackles the issue of allocating students to industry placements. In the model proposed in this work, each company can offer one or more placements, and companies interview students to define their preferences. The model aims to optimize company satisfaction but does not take into account student preferences. To the best of our knowledge, the proposal presented in this paper is the first to address an optimization model for SIPA problems that considers different types of preferences associated with students and companies. Unlike other works, we model preferences based on competences. For this, we propose a multi-objective integer linear model.

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2 INTEGER LINEAR PROGRAMMING MODEL

Let us define the set of students to be allocated by an educational institution as $\mathcal{S} = \{1, \ldots, s\}$. Let $\mathcal{P} = \{1, \ldots, p\}$ denote the placements that the education center has available for students. In addition to this, $\mathcal{B} = \{1, \ldots, b\}$ represents the set of companies that offer any placement for the educational center. Each placement $j \in \mathcal{P}$ is associated with a given company. $P_k \subset \mathcal{P}$ represents the set of placements that are offered by company k. Each company $k \in \mathcal{B}$ has agreed with the educational center a minimum and maximum number of students to be allocated (q_k, Q_k). Furthermore, $\mathcal{I} = \{1, \ldots, n\}$ represent n different sets of placements that must be allocated together. Finally, $C = \{1, \ldots, c\}$ refer to the set of competences that are relevant for the assignment of students to placements. The proposed model assumes that each placement $j \in \mathcal{P}$ has associated a vector of competence relevance $\Phi_j =< \phi_{j1}, \ldots, \phi_{jc} >$, where each ϕ_{jl} represents the importance of competence of $l \in C$ for the development of placement j. Similarly, a student i has two vectors associated with competences. $\Psi_i =< \psi_{i1}, \ldots, \psi_{ic} >$, where each ψ_{il} represents the current development of competence $l \in C$ by student i, and $\Theta_i =< \theta_{i1}, \ldots, \theta_{ic} >$, where each θ_{il} represents the preference of student i for developing competence l during her placement. Given these definitions, we formally define the utility reported by placement j to student i as $\tau_{ij} = w_{id} \times (1 - d_{ij}) + w_{ic} \times (\Phi_j \cdot \Theta_i)$, where w_{id} is the importance given by student i to the distance from her home to the allocated placement, w_{ic} is the importance

given by the student to the competences to be developed in the placement, and d_{ij} is the distance between the student's home and the workplace of the placement. The formula also includes the dot product of Φ_j and Θ_j as an average of the preferences of students with respect to competences weighted by the importance of each competence for the placement. The utility reported by student i to placement j, is defined as $\eta_{ij} = \Phi_j \cdot \Psi_i$ where the formula represents an average of the student's competences weighted by the importance of each competence for the placement. Accordingly, we formalize a multi-objective integer linear programming model.

$$\max \quad \frac{\alpha}{|\mathcal{S}|}\sum_{i=1}^{s}\sum_{j=1}^{p}\tau_{ij}\times x_{ij} + \frac{\beta}{|\mathcal{S}|}\sum_{i=1}^{s}\sum_{j=1}^{p}\eta_{ij}\times x_{ij} \tag{1a}$$

s.t.

$$\sum_{j=1}^{p} x_{ij} = 1 \qquad \forall i \in \mathcal{S}, \tag{1b}$$

$$\sum_{i=1}^{s} x_{ij} \leq 1 \qquad \forall j \in \mathcal{P}, \tag{1c}$$

$$\sum_{i=1}^{s}\sum_{j\in P_k} x_{ij} \leq Q_k \qquad \forall k \in \mathcal{B}, \tag{1d}$$

$$\sum_{i=1}^{s}\sum_{j\in P_k} x_{ij} \geq q_k \qquad \forall k \in \mathcal{B}, \tag{1e}$$

$$\left(\sum_{i=1}^{s}\sum_{j\in I} x_{ij}\right) - |I| \times \delta_I \leq 0 \qquad \forall I \in \mathcal{I}, \tag{1f}$$

$$|I| \times \delta_I - \sum_{i=1}^{s}\sum_{j\in I} x_{ij} \leq 0 \qquad \forall I \in \mathcal{I} \tag{1g}$$

where $x_{ij} \in \{0,1\}$ are binary decision variables that represent if student i is allocated to placement j, $\delta_I \in \{0,1\}$ are binary decision variables that indicate if interdependent placements $I \in \mathcal{I}$ are allocated or not. The multi-objective aspect of the problem is modeled by means of a weighted sum of the two objectives. Each objective is represented by one of the two terms in the function. The first term represents the average utility of allocations from the students' perspective, and it is related to the overall satisfaction of students with the allocation. The second term represents the average utility reported to businesses by the proposed allocations, and it is related to the overall satisfaction of companies with the allocation. In addition to this, constraints 1b guarantee that each student is allocated to exactly one placement, constraints 1c ensure that each placement is allocated at most once, and constraints 1d and 1e guarantee that each company is allocated a number of students within the minimum and maximum agreed number of students. Constraints 1f and 1g force placements with interdependencies to be allocated together in case they are allocated.

3 EXPERIMENTS

In order to study the performance of our model under different solvers, we selected two non-commercial (SCIP and CBC) and one commercial solver (Gurobi). Despite non-comercial solvers usually being slower than commercial ones, we are interested in analyzing their performance as many educational institutions may not have access to commercial software for non-research or non-educational purposes. We employed 120 problem instances generated by varying the number of students, the number of placements available, the students' and businesses' locations, and the distribution and number of competences for students' and placements. Each problem instance and configuration was solved 10 different times to capture stochastic variations in the computation time.

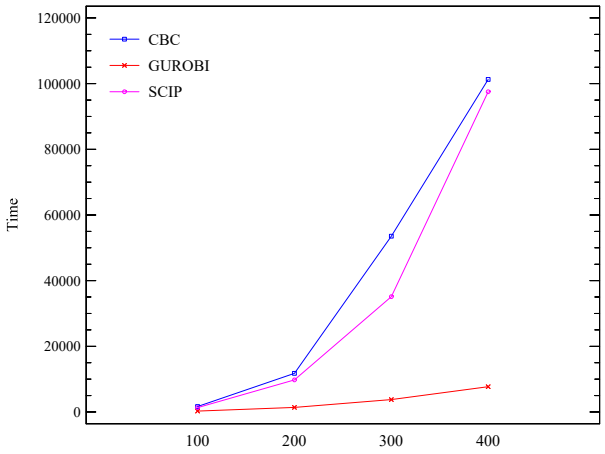


Figure 1: Mean execution time for each solver according to the number of students to be allocated

Figure 1 shows the increasing tendency of the mean execution time and the number of students. It can be highlighted that the mean execution time grows in a possibly exponential way. Gurobi has a slower growth compared to CBC and SCIP, being the fastest of the three solvers and showing consistent behavior across problem instances. Despite showing a good performance, not all schools and educational institutions may have access to commercial solvers. Hence, they may need to employ non-commercial solvers, which have a faster growth for the mean execution time and whose mean execution times may result in unacceptable for larger problem instances. Consequently, one conclusion that can be drawn from these experiments is the necessity for alternative methods to solve the problem when one faces larger problem instances.

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